ocean in the form of heat. During an El Niño water temperatures in the Pacific Ocean may rise on average 3 - 5 degrees above average. This happens as the water in seas around Indonesi

Topography

Currents and tides influence topography, as does temperature. Water expands as it gets warmer, and the lack of cold water dependent nutrients make it less dense. This expanded, less dense water results in a rise in sea level, observable from space. Ocean surface height may rise as much as 6 to 13 inches above normal in some ocean regions during an El Niño.

Winds

The QuikSCAT satellite tracks vector winds, and the Jason and Topex/Poseidon missions track their effects. Typically, the Pacific trade winds blow from east to west, dragging the sunlit warm waters westward where they accumulate in

how the physical properties of the ocean interact to create climate patterns. Scientists will continue to create models that combine data allowing them to better understand and predict complex ocean processes.

ability to study the El Niño climate pattern and create models to simulate conditions, has helped us better predict its impact on our climate and weather.

El Niño is only one of the climate anomalies that scientists have observed;others include the North Atlantic Oscillation, the Atlantic Intertropical Convergence Zone oscillation, the Pacific Decadal Oscillation. Together with El Niño these systems are believed to be responsible for well over fifty percent of the climate variability on Earth. As models are created to simulate the individual patterns they will be combined to create a global model of climate change. If scientists get to the point to where they understand all of these climate cycles they may be able to predict major weather patters months in advance.

April 25, 2023

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